

DreamCampus — Physics Detailed Formula Book

Chapter-wise formulas, conditions and revision reminders

Mechanics

Topic	Formula(s)	Reminder
Kinematics	$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$	Constant acceleration
Projectile	$R = \frac{u^2 \sin 2\theta}{g}$; $H = \frac{u^2 \sin^2 \theta}{(2g)}$; $T = \frac{2u \sin \theta}{g}$	Same launch/landing level
Newton's laws	$F_{\text{net}} = ma$; weight = mg ; $f_k = \mu_k N$	Use free-body diagrams
Work & energy	$W = Fd \cos \theta$; $K = \frac{1}{2}mv^2$; $U = mgh$; $P = W/t$	Check force/displacement angle
Momentum	$p = mv$; $J = \Delta p$	Impulse-momentum theorem
Circular motion	$a_c = v^2/r = \omega^2 r$; $F_c = mv^2/r$	Inward net force
Rotation	$\tau = rF \sin \theta$; $L = I\omega$; $K_{\text{rot}} = \frac{1}{2}I\omega^2$; $v = R\omega$	Rolling without slipping
Gravitation	$F = G\frac{m_1 m_2}{r^2}$; $v_{\text{orbit}} = \sqrt{GM/r}$; $v_{\text{escape}} = \sqrt{2GM/R}$	Circular orbit / escape

Properties & Thermodynamics

Topic	Formula(s)	Reminder
Pressure & density	$P = F/A$; $\rho = m/V$; $F_b = \rho Vg$	Idealized fluid relations
Fluid flow	$A_1 v_1 = A_2 v_2$	Steady incompressible flow
Bernoulli	$P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$	Along a streamline in ideal flow
Heat	$Q = mc\Delta T$; $Q = mL$	Specific heat / latent heat
Ideal gas	$PV = nRT$	n = number of moles
First law	$\Delta Q = \Delta U + W$	W positive when done by system in this convention

Oscillations & Waves

Topic	Formula(s)	Reminder
SHM	$a = -\omega^2 x$; $x = A \cos(\omega t + \phi)$	Restoring acceleration
Spring	$T = 2\pi\sqrt{m/k}$; $\omega = \sqrt{k/m}$	Ideal spring
Pendulum	$T \approx 2\pi\sqrt{L/g}$	Small-angle approximation
Wave	$v = f\lambda$; $\omega = 2\pi f$; $k = 2\pi/\lambda$	Basic progressive wave

Electrostatics

Topic	Formula(s)	Reminder
Coulomb	$F = kq_1 q_2 / r^2$	$k = 1/(4\pi\epsilon_0)$
Field	$E = F/q$; $E_{\text{point}} = kq/r^2$	Vector quantity

Potential	$V_{\text{point}} = kq/r$	Scalar quantity
Capacitor	$C = Q/V$; $U = \frac{1}{2}CV^2 = Q^2/(2C)$	Energy storage

Current Electricity

Topic	Formula(s)	Reminder
Current	$I = Q/t$ (average); $I = dQ/dt$	Charge flow rate
Ohm's law	$V = IR$	For ohmic behavior
Power	$P = VI = I^2R = V^2/R$	Electrical power
Resistors	$R_{\text{series}} = \Sigma R$; $1/R_{\text{parallel}} = \Sigma(1/R)$	Ideal combinations

Magnetism & EMI

Topic	Formula(s)	Reminder
Magnetic force	$F = qvB\sin\theta$; $F_{\text{wire}} = BIL\sin\theta$	Direction by right-hand rules
Flux	$\Phi = BA\cos\theta$	Uniform field/simple surface
Faraday-Lenz	$\mathcal{E} = -d\Phi/dt$	Minus sign represents Lenz's law
AC reactance	$X_L = \omega L$; $X_C = 1/(\omega C)$	Ideal inductor/capacitor

Optics

Topic	Formula(s)	Reminder
Refraction	$n_1 \sin i = n_2 \sin r$	Snell's law
Critical angle	$\sin C = n_2/n_1$ for $n_1 > n_2$	Total internal reflection
Lens	$1/f = 1/v + 1/u$	Use adopted sign convention
Mirror	$1/f = 1/v + 1/u$	Use adopted sign convention
Power	$P = 1/f$ (f in metres)	Lens power in dioptres
Wave optics	$\beta = \lambda D/d$ (YDSE fringe width)	Ideal double-slit geometry

Modern Physics

Topic	Formula(s)	Reminder
Photon	$E = hf = hc/\lambda$	Photon energy
Photoelectric	$K_{\text{max}} = hf - \phi$	Threshold frequency $f_0 = \phi/h$
de Broglie	$\lambda = h/p$	Matter waves
Mass-energy	$E = mc^2$	Rest-energy relation
Radioactivity	$N = N_0 e^{(-\lambda t)}$; $T_{1/2} = 0.693/\lambda$	Exponential decay

Formula revision checklist

For every formula learn: meaning of symbols, SI units, conditions/assumptions, vector or scalar nature, and the sign convention. Do not memorize a formula without knowing when it applies.